

Business Case Study

OLA - Ensemble Learning-Heena

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Colab Link: <https://drive.google.com/file/d/10BYNm51YvDi61mQKqPkwM-bwrIDF7wWM/view?usp=sharing>

Problem Statement

Recruiting and retaining drivers is seen by industry watchers as a tough battle for Ola. Churn among drivers is high and it's very easy for drivers to stop working for the service on the fly or jump to Uber depending on the rates.

As the companies get bigger, the high churn could become a bigger problem. To find new drivers, Ola is casting a wide net, including people who don't have cars for jobs. But this acquisition is really costly. Losing drivers frequently impacts the morale of the organization and acquiring new drivers is more expensive than retaining existing ones.

You are working as a data scientist with the Analytics Department of Ola, focused on driver team attrition. You are provided with the monthly information for a segment of drivers for 2019 and 2020 and tasked to predict whether a driver will be leaving the company or not based on their attributes like

- Demographics (city, age, gender etc.)
- Tenure information (joining date, Last Date)
- Historical data regarding the performance of the driver (Quarterly rating, Monthly business acquired, grade, Income)

Dataset

Dataset Link: [ola_driver.csv](#)

Column Profiling:

1. MMMM-YY : Reporting Date (Monthly)
2. Driver_ID : Unique id for drivers
3. Age : Age of the driver
4. Gender : Gender of the driver – Male : 0, Female: 1
5. City : City Code of the driver
6. Education_Level : Education level – 0 for 10+ ,1 for 12+ ,2 for graduate
7. Income : Monthly average Income of the driver
8. Date Of Joining : Joining date for the driver
9. LastWorkingDate : Last date of working for the driver
10. Joining Designation : Designation of the driver at the time of joining
11. Grade : Grade of the driver at the time of reporting
12. Total Business Value : The total business value acquired by the driver in a month (negative business indicates cancellation/refund or car EMI adjustments)
13. Quarterly Rating : Quarterly rating of the driver: 1,2,3,4,5 (higher is better)

Concept Used:

- Ensemble Learning- Bagging
- Ensemble Learning- Boosting
- KNN Imputation of Missing Values
- Working with an imbalanced dataset

Problem Statement:

Ola is facing high driver attrition. Replacing drivers is costly, so the objective is to identify drivers who are likely to leave and understand the factors associated with attrition.

Dataset contains **2,381 unique drivers** after aggregation, compared with **19,104 monthly records** originally.

Insight :

The original data is monthly, so the same driver appears multiple times. Aggregating at the driver level is therefore necessary to make the prediction unit:

One driver = one observation

Business recommendation:

Ola should build a **driver-level churn-risk system**, rather than treating each monthly record as an independent driver.

Shape of Original Dataset:

Original dataset:

- **19,104 rows**
- **14 columns**
- **2,788 unique Driver IDs approximately**
- Monthly observations over the reporting period.

Insight :

The large number of rows does **not mean Ola has 19,104 drivers**.

It represents repeated monthly observations of drivers.

Business recommendation:

For attrition prediction, aggregate monthly information into driver-level features such as:

- Average income
- Maximum income
- Total business value
- Average rating
- Maximum rating
- Rating improvement
- Income improvement
- Tenure

Number of monthly reports

Data Types:**Insight :**

Date variables contain valuable information that cannot be effectively used as raw strings.

For example:

Dateofjoining

can be transformed into:

- Tenure
- Year of joining
- Number of months with Ola

Business recommendation:

For future production implementation, derive more temporal features such as:

- Months since joining
- Recent rating change
- Recent income change
- Recent business-value trend

Missing Value Analysis:

Insights:

The **91.54% missing LastWorkingDate is not a data-quality problem.**

It occurs because:

If a driver is still working, there is no LastWorkingDate.

Therefore:

LastWorkingDate missing → driver has not left

LastWorkingDate present → driver has left

Business recommendation:

Do **not** impute LastWorkingDate.

Age Missing Values:

Insight :

This is a very small proportion of the dataset.

It is unlikely to materially affect the analysis, but the missing values should still be handled systematically.

Business recommendation:

KNN imputation is reasonable here because age can be estimated using similar drivers based on other numerical characteristics.

Gender Missing Values:

Insight :

The amount of missing gender information is extremely small.

Business recommendation:

There is no need to remove the entire rows.

Your use of imputation avoids unnecessary loss of data.

Original Target Distribution:
Original Target Distribution

training data has:

- Class 1 → 1,292
- Class 0 → 612

Business recommendation:

The original monthly dataset was highly imbalanced at the record level. After aggregating the monthly observations into one observation per driver, the class distribution changed because the prediction unit became the driver rather than the monthly record.

Driver-Level Aggregation:

Insight :

This is one of the strongest parts of your analysis.

Instead of looking at a driver's individual monthly record, you summarize their **overall relationship and performance history with Ola**.

Business recommendation:

For a production system, I would improve this further by using **recent behaviour**.

For example:

- Last 3-month income
- Last 3-month rating
- Rating trend
- Business-value trend
- Income trend

This would make the model more suitable for **early churn prediction**.

Age Distribution:

Insight :

The driver population is primarily working-age adults, with a median age of about 33 years.

The range of 21–58 does not immediately suggest impossible values.

Business recommendation:

Age should be retained as a predictive variable, but Ola should **not make retention decisions based on age alone**.

Instead, age should be considered together with:

- Income
- Rating
- Grade
- Tenure
- Business performance

Income Distribution:

Insight :

The median income of drivers who left is approximately:

₹12,002 lower

than that of retained drivers.

Percentage difference relative to retained median:18.9%

So the median income of churned drivers is about **19% lower** than retained drivers in your dataset.

Business Interpretation:

Lower income is associated with higher observed attrition.

This does **not prove that low income causes attrition**, but it is a strong signal worth investigating.

Business recommendation:

Ola could:

- Review driver earning potential
- Identify drivers with persistently low income
- Provide targeted incentives
- Investigate city-wise earning differences
- Improve demand allocation
- Monitor earnings during the first few months

Rating Increase:

Insight :

Among drivers whose rating **did not increase**, about **80.5% were classified as having left**.

Among drivers whose rating **increased**, the observed attrition proportion was about **44.0%**.

That's a substantial difference.

Business interpretation

A lack of performance improvement appears to be strongly associated with attrition.

Business recommendation:

Ola should create a:

Driver Performance Early-Warning System : This is much better than waiting until the driver actually leaves.

Drivers whose ratings did not show an increase had substantially higher observed attrition in this dataset.

Income Increase:**Insight :**

Among drivers whose income increased, only about **6.82%** were classified as having left.

Among drivers whose income did not increase, approximately **69.02%** were classified as having left.

Business interpretation

Income growth is strongly associated with retention in your dataset.

Business recommendation:

This should be one of your **top business recommendations**.

Ola could develop:

"Driver Earnings Growth Program"

Monitor:

- Income trend
- Number of trips
- Incentive achievement
- Business value
- City demand

If income remains stagnant for multiple periods:

Flag the driver as a potential churn risk.

Then Ola can investigate whether:

- Demand is insufficient
- Incentives are inadequate
- Driver is inactive
- Competition is offering better earnings
- Operational issues exist

Tenure:**Insight :**

Drivers who left had somewhat shorter median tenure.

This suggests that **early tenure may be an important period for retention**.

Business recommendation:

Introduce a:

First 6-month Driver Retention Program

For example:

Month 1

- Onboarding support

Month 2–3

- Earnings monitoring

Month 3–4

- Performance feedback

Month 5–6

- Churn-risk assessment

This is especially relevant because the difference in median tenure is not enormous; therefore, tenure should be combined with income/rating rather than used alone.

Grade vs Attrition:

Insight :

There is a strong relationship between lower grades and higher observed attrition.

Grade 1 has approximately **80.4% attrition**, compared with approximately **50–54%** for grades 3–5.

Observation

Grade 5 does not have the lowest attrition.

Therefore, we should **not simply say "higher grade always means lower attrition."**

The relationship is not perfectly monotonic.

Business recommendation:

For lower-grade drivers:

- Identify performance barriers
- Provide coaching
- Review incentive eligibility
- Provide operational support
- Monitor rating improvement

But don't automatically assume that every low-grade driver will leave.

City-Level Attrition:

Insight :

There is substantial variation in observed attrition across cities.

For example:

C13 → approximately 81.7%

versus cities with lower observed rates.

Business interpretation

Location appears to be an important segmentation variable.

However, **city itself may not be the underlying cause.**

Possible explanations include:

- Different earning potential
- Demand/supply imbalance
- Competition
- Incentive structures
- Driver demographics
- Operational conditions

Business recommendation:

Ola should perform a:

City-wise Driver Retention Analysis

For high-attribution cities, compare:

- Median income
- Income growth
- Rating
- Business value
- Tenure
- Driver grade
- Number of trips
- Incentives

Then design **city-specific retention programs.**

Total Business Value:

Insight :

Business value captures driver contribution over time.

A driver with consistently strong business contribution may represent a valuable resource to retain.

Business recommendation:

Ola can create a **Value × Churn Risk matrix.**

Correlation Analysis:

Insight :

As rating improvement increases, observed attrition tends to decrease.

This is consistent with your crosstab:

No rating increase → 80.49% attrition

Rating increase → 43.99% attrition

Business recommendation:

Track **performance trajectory**, not just current performance.

This is an important analytical lesson:

Trend can be more informative than a single snapshot.

Class Imbalance:

Insight :

SMOTE created a balanced training dataset.

Business recommendation:

This is appropriate because missing the churn class can be costly to the business.

Random Forest — Results:

Business interpretation

The model correctly identified:

309 out of 324 actual churners

So it missed:

15 churners

That's a relatively small number of false negatives.

Business recommendation:

Random Forest is already a strong baseline for this business problem.

Gradient Boosting — Results:

Insight :

Compared with Random Forest:

- Accuracy improved
- Precision improved
- F1 improved slightly
- ROC-AUC improved

But recall decreased slightly.

Business meaning

Gradient Boosting gives a slightly different balance between:

catching churners and avoiding unnecessary retention interventions.

Tuned Random Forest:

Insight :

Tuning improved some classification metrics slightly, but:

ROC-AUC actually decreased from 0.9639 to 0.9627.

Therefore, hyperparameter tuning did **not universally improve the model**.

Hyperparameter tuning does not guarantee improvement on every metric. Therefore, I evaluated the tuned model on the same holdout test set rather than assuming that tuning automatically produced a better model.

Tuned Gradient Boosting:

Insight :

Interestingly:

- Accuracy decreased compared with basic Gradient Boosting.
- F1 decreased slightly.
- ROC-AUC increased from **0.9677** → **0.9688**.

So tuning improved the ranking/discrimination capability measured by ROC-AUC, but not the threshold-based classification metrics.

Overall Model Results:

The models provide broadly similar performance. Gradient Boosting gives the highest accuracy and F1 among the untuned models, while tuned Gradient Boosting gives the highest ROC-AUC. Random Forest provides the highest recall for the churn class among the tested models.

ROC-AUC:**Insight :**

All four models demonstrate strong ability to distinguish between the two target classes on holdout dataset.

The highest observed ROC-AUC is:

0.9688 for tuned Gradient Boosting

Business recommendation:

For a production churn-risk system, ROC-AUC alone should not determine the final operating model.

Ola should also consider:

- Recall
- Precision
- Cost of contacting a driver
- Cost of losing a driver
- Retention incentive cost

Year of Joining:**Business recommendation:**

Instead of acting directly on "year of joining", investigate **cohort-level retention**.

Quarterly Rating:**Insight :**

Driver performance appears to have a meaningful relationship with attrition.

This is consistent with your earlier finding:

Rating improvement → substantially lower observed attrition.

Business recommendation:

Ola should monitor rating trajectory monthly/quarterly and trigger intervention when performance stagnates.

Tenure:**Insight :**

How long a driver has remained with Ola carries substantial information about churn risk.

Combined with your median tenure finding:

Retained → 0.58 years

Churned → 0.48 years

there is evidence that early tenure deserves special attention.

Business recommendation:

Develop a dedicated **new-driver retention program**, particularly during the first 6–12 months.

Number of Reportings:**Insight :**

The number of months for which the driver has reporting information is predictive.

This is closely related to driver engagement/tenure.

Business recommendation:

For a true **early-warning model**, be careful about using features that are only known after substantial time has passed.

Total Business Value:**Insight :**

Driver business contribution is an important predictor.

Business recommendation:

Ola should combine:

Business value × churn probability

to prioritize retention.

A high-value driver with high churn probability should receive more urgent attention than a low-value driver with the same predicted risk.

Income:

Business recommendation:

Income should be monitored not just as an absolute amount but as:

Income trajectory

Final Business Recommendations:

Recommendation 1 — Driver Churn Early-Warning System

Use the model to assign every driver a:

Churn Probability

For example:

0–30% Low Risk
30–60% Medium Risk
60–100% High Risk

These thresholds should be validated using business costs rather than treated as universal.

Recommendation 2 — Focus on Income Growth

Your strongest business finding is:

Drivers with income increase had only **6.82% observed attrition**, compared with **69.02%** among drivers without income increase.

Therefore:

- Monitor income trends
- Identify stagnant earnings
- Review incentives
- Improve demand allocation
- Investigate city-level earning differences

Recommendation 3 — Monitor Rating Trends

Drivers whose ratings did not increase had approximately:

80.49% observed attrition

versus approximately:

43.99% for drivers whose rating increased.

Therefore:

Don't wait for a driver to reach a poor rating.

Monitor whether the rating is **improving or stagnating**.

Recommendation 4 — Special Attention to New Drivers

Because churned drivers have a lower median tenure:

0.48 years vs 0.58 years

Ola should strengthen:

- First 3-month onboarding
- First 6-month support
- Earnings monitoring
- Performance feedback
- Incentive communication

Recommendation 5 — City-Specific Retention

Cities such as:

- C13
- C17
- C23
- C2

show very high observed attrition rates in your analysis.

Instead of giving the same incentive everywhere, Ola should investigate **why** these cities have higher attrition.

The analysis indicates that driver retention is strongly associated with income growth, rating improvement, tenure, business performance and city. In particular, drivers with increasing income and improving ratings showed substantially lower observed attrition in this dataset. The ensemble models were able to identify churn with approximately 0.96–0.97 ROC-AUC, which suggests that these features contain useful predictive information. Ola could use such a model as an early-warning system and combine the risk score with driver value and operational factors to prioritize retention interventions.